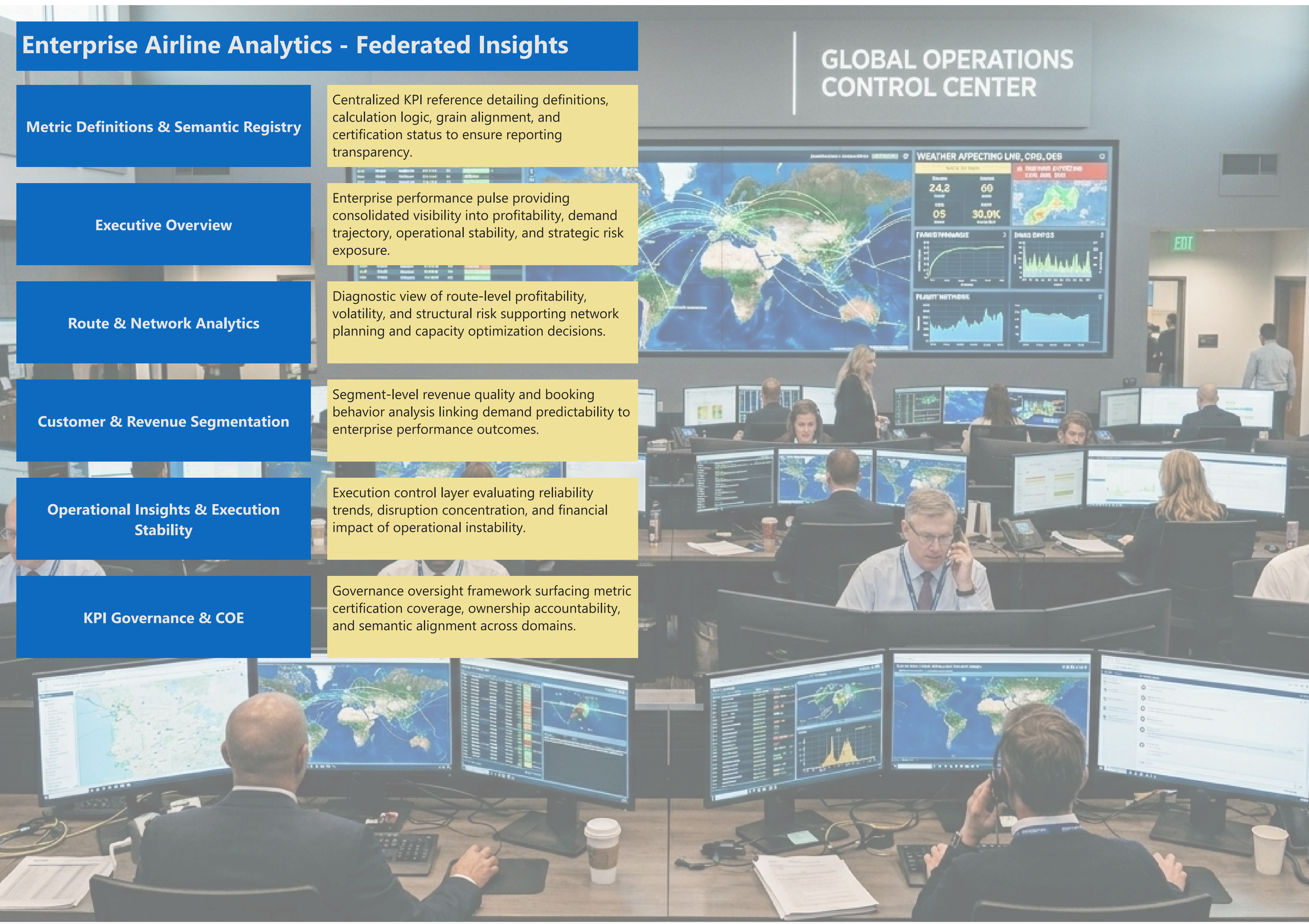
Enterprise Airline Analytics - Federated Insights

Metric Definitions & Semantic Registry	Centralized KPI reference detailing definitions, calculation logic, grain alignment, and certification status to ensure reporting transparency.
Executive Overview	Enterprise performance pulse providing consolidated visibility into profitability, demand trajectory, operational stability, and strategic risk exposure.
Route & Network Analytics	Diagnostic view of route-level profitability, volatility, and structural risk supporting network planning and capacity optimization decisions.
Customer & Revenue Segmentation	Segment-level revenue quality and booking behavior analysis linking demand predictability to enterprise performance outcomes.
Operational Insights & Execution Stability	Execution control layer evaluating reliability trends, disruption concentration, and financial impact of operational instability.
KPI Governance & COE	Governance oversight framework surfacing metric certification coverage, ownership accountability, and semantic alignment across domains.



METRIC DEFINITIONS & SEMANTIC REGISTRY

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For questions & inquiries, contact Anil Jacob at Aniljacobs@gmail.com, or visit my GitHub
This project repo is available at the following link: https://github.com/AwesomeAnil/Federated-Insights-Case-Study

EXECUTIVE OVERVIEW - YEAR 2024

Close Panel

Hide Yellow box commentary

Show Yellow box commentary

Select a Year:

- ○ 2022
- ○ 2023
- ◉ 2024

Select a route:

- ☐ DXB-BOM
- ☐ DXB-JFK
- ☐ DXB-LHR
- ☐ DXB-SIN
- ☐ DXB-SYD

Select a fare class:

- ☐ Economy
- ☐ Premium Economy
- ☐ Business
- ☐ First

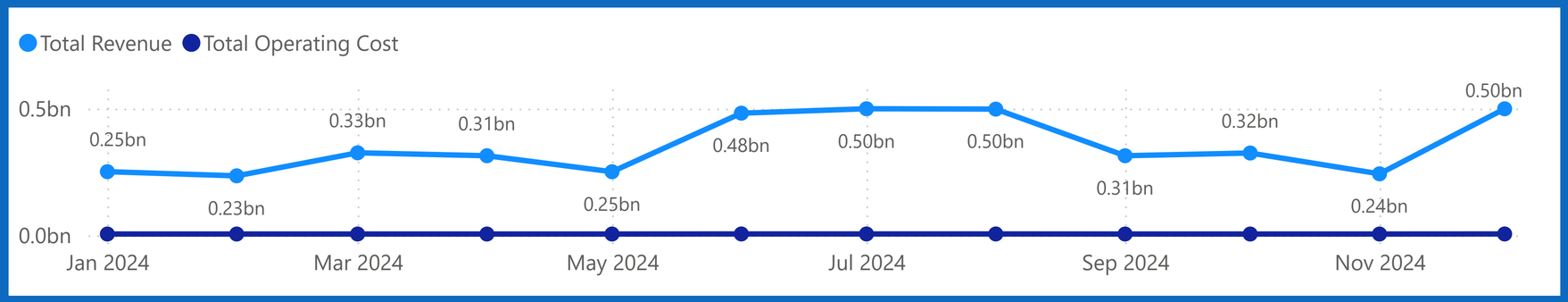
Select a segment:

- ☐ Business Traveller
- ☐ Corporate Contract
- ☐ Leisure Traveller
- ☐ Loyalty Gold

Executive KPIs

4.2bn	4.2bn	98.6%	89.7%	-0.13
Total Revenue	Route Profit	Profit Margin %	Revenue Forecast A...	Demand Momentu...

Financial Performance & Trends



Demand & Operational Signals

79.9%	39.3%
Load Factor %	On-Time Departure %
4.4%	**71.83**
Cancellation Rate %	Average of Tickets S...

Risk & Exposure

Route_ID	Route Risk Score	Profit Margin %	Load Factor %
DXB-SYD	2.69	98.3%	79.9%
DXB-JFK	2.64	98.5%	80.0%
DXB-LHR	2.62	98.7%	79.8%
DXB-BOM	2.61	98.8%	80.0%
Total	**2.63**	**98.6%**	**79.9%**

0.44	2.63
Fuel Cost Ratio	Route Risk Score

Executive KPI's

Consolidated view of revenue growth, profitability, forecast reliability, and forward demand momentum to assess overall business health and trajectory.

Financial Performance Trends:

Illustrates revenue progression, operating cost pressure, and margin resilience across seasonal demand cycles and fuel volatility impacts.

Demand and Operational Signals:

Surfaces booking acceleration trends and operational stability indicators that influence near-term revenue realization and customer reliability.

Risk & Exposure:

Highlights route-level volatility, margin sensitivity, and operational disruption risk to support proactive network and cost management decisions.

ROUTE ANALYTICS (NETWORK PLANNING) - YEAR 2024

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Show Yellow box commentary

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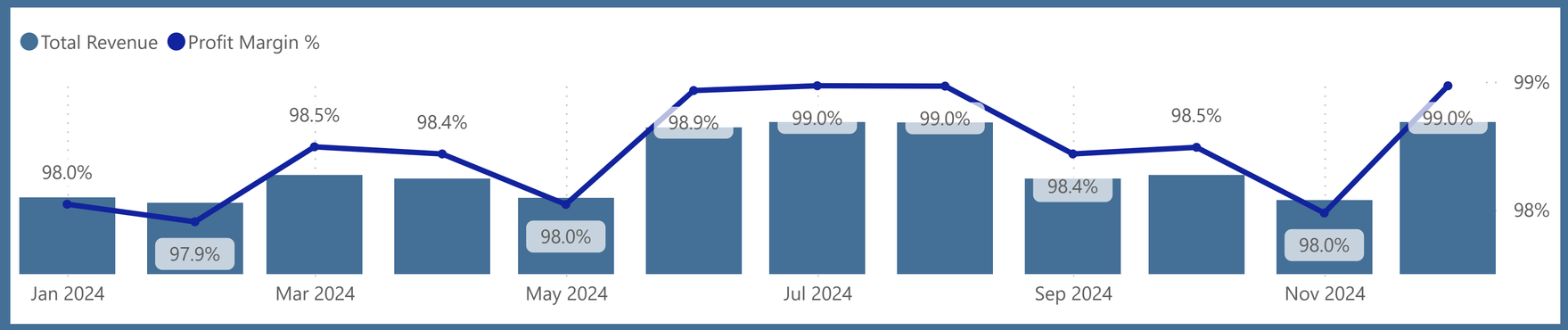
Route Health Snapshot

4.2bn	98.6%	79.9%	2.63	4.99M
Total Revenue	Profit Margin %	Load Factor %	Route Risk Score	Cost per Route

Route Health Snapshot

Provides a consolidated health view of the selected route, combining profitability, demand strength, and structural risk to assess strategic viability.

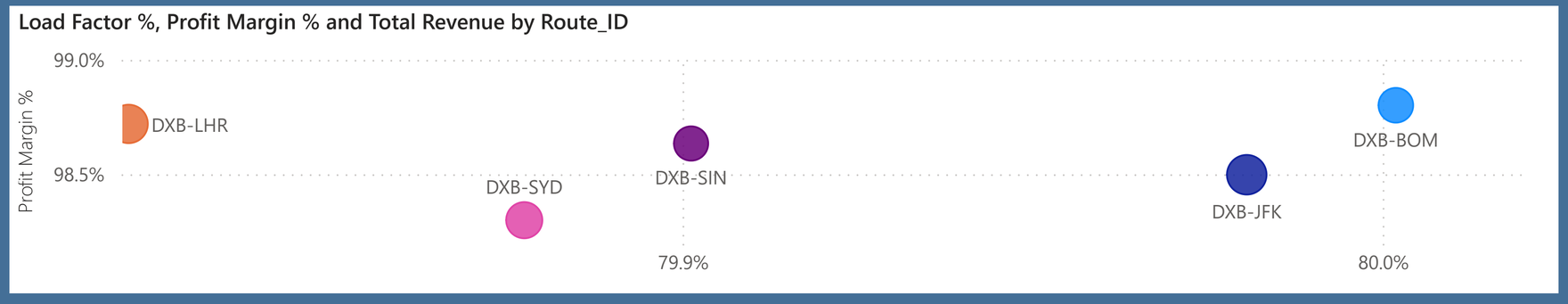
Route Revenue & Margin Trend



Route Revenue & Margin Trend

Reveals whether profitability is improving, deteriorating, or structurally constrained by cost pressure.

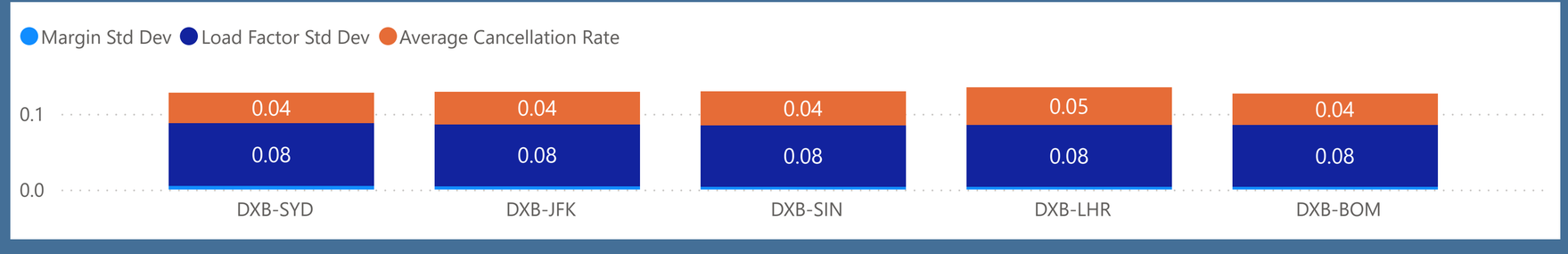
Performance Quadrant



Performance Quadrant

Categorizes network performance by load strength and margin resilience to guide capacity and pricing decisions.

Risk & Volatility breakdown



Risk & Volatility

Deconstructs the volatility drivers influencing route-level risk exposure across margin stability, demand variability, and operational disruption.

CUSTOMER & COMMERCIAL SEGMENTATION- YEAR 2024

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- ◉ 2024

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Select a segment:

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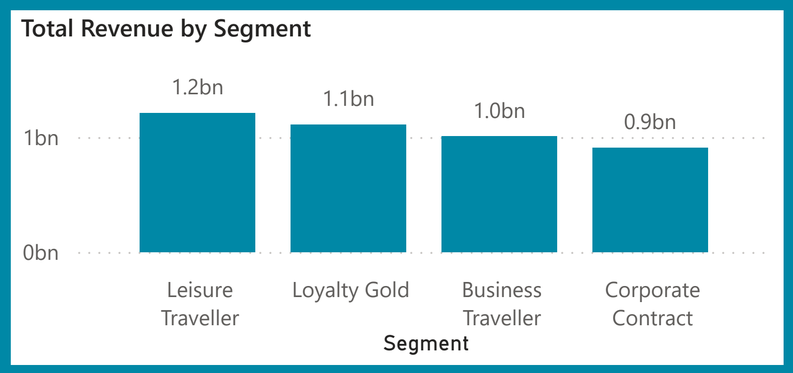
- ☐ Leisure Traveller
- ☐ Loyalty Gold

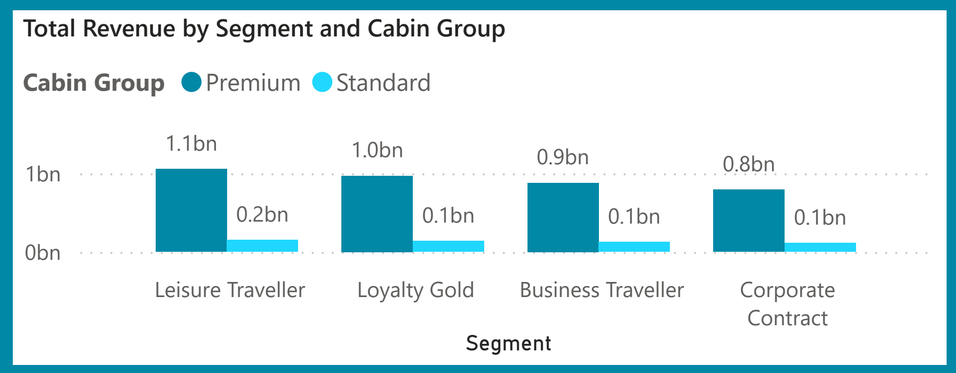
Segment Performance Snapshot

4.2bn	2,013	87.3%	2M	45
Total Revenue	Yield per Passenger	Premium Cabin Reve...	Total Passengers	Advance Booking Wi...

Segment Performance Snapshot:
Consolidates segment-level revenue contribution, yield quality, and booking behavior to assess structural demand value.

Revenue & Mix Analysis

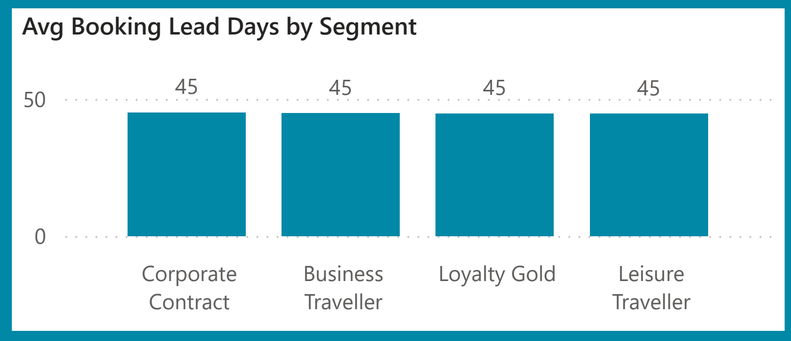


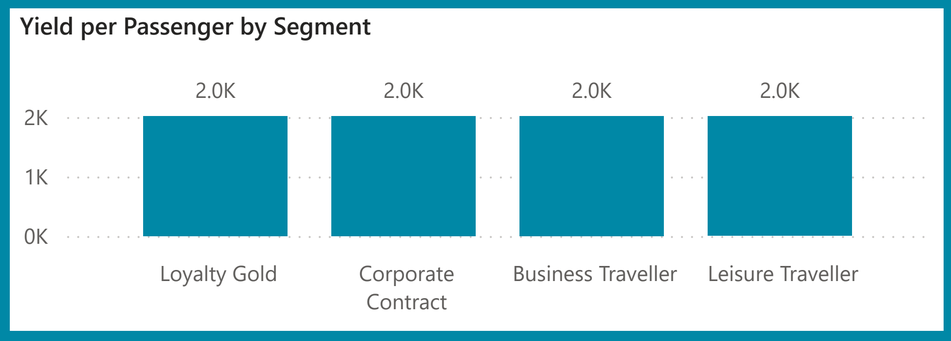


Revenue & Segment Mix Analysis:

Evaluates revenue concentration and premium mix distribution to highlight structurally higher-margin customer groups.

Booking Behavior & Yield Analysis

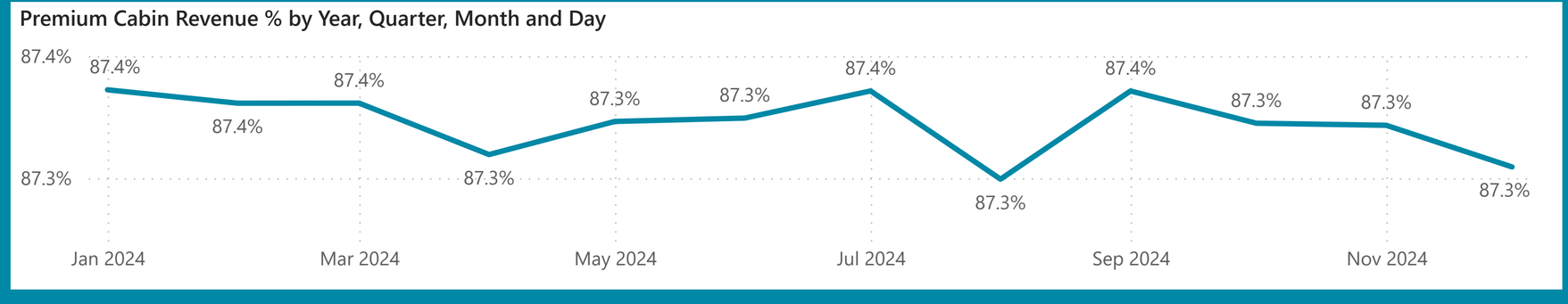




Booking Lead Days:

Reveals advance booking behavior differences across segments to assess demand visibility and forecasting reliability.

Premium Demand Trend



Premium Booking Trend:

Tracks evolution of premium revenue share across segments to detect long-term shifts in customer mix and profitability drivers.

OPERATIONAL INSIGHT (EXEC CONTROL VIEW) - YEAR 2022

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Select a Year:

- (Blank)
- 2022
- 2023
- 2024

Select a route:

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- DXB-SIN
- DXB-SYD

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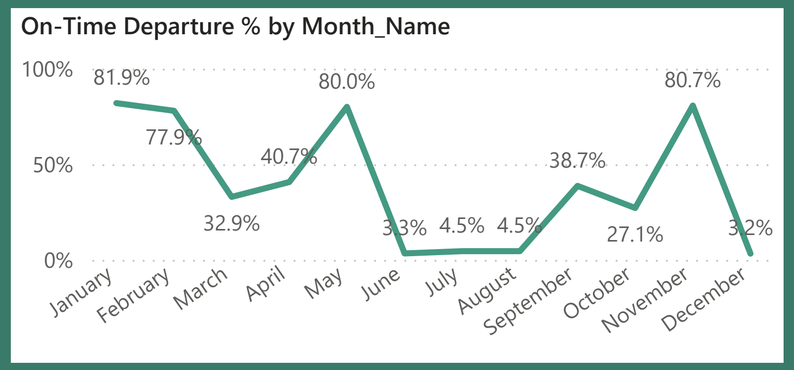
- Loyalty Gold

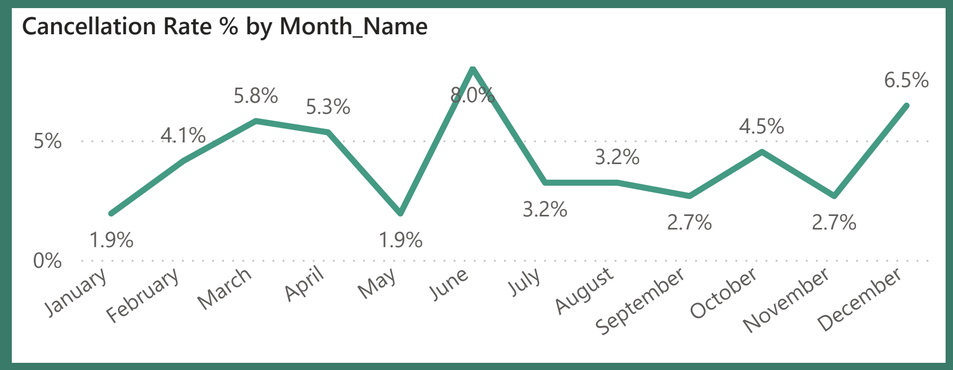
Operational Stability Snapshot

39.4%	4.2%	79.8%	2M	4.2bn
On-Time Departure %	Cancellation Rate %	Load Factor %	Total Passengers	Total Revenue

Operational Stability Snapshot:
Provides a consolidated view of execution reliability and disruption exposure to assess overall operational stability

Delay & Cancellation Trends

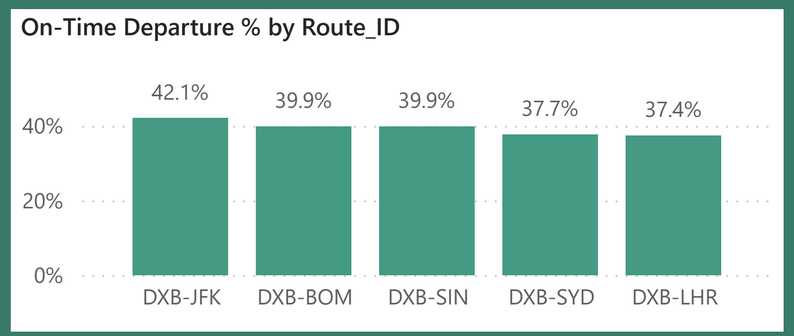


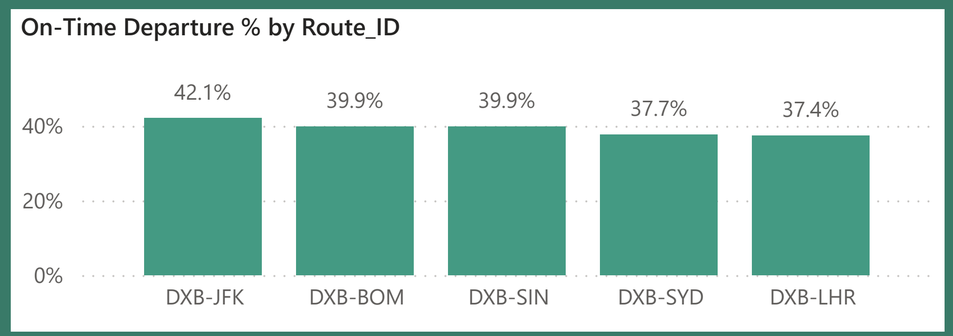


Delay & Cancellation Trends:

Reveals structural reliability patterns over time to distinguish temporary volatility from sustained operational weakness.

Operational Concetration

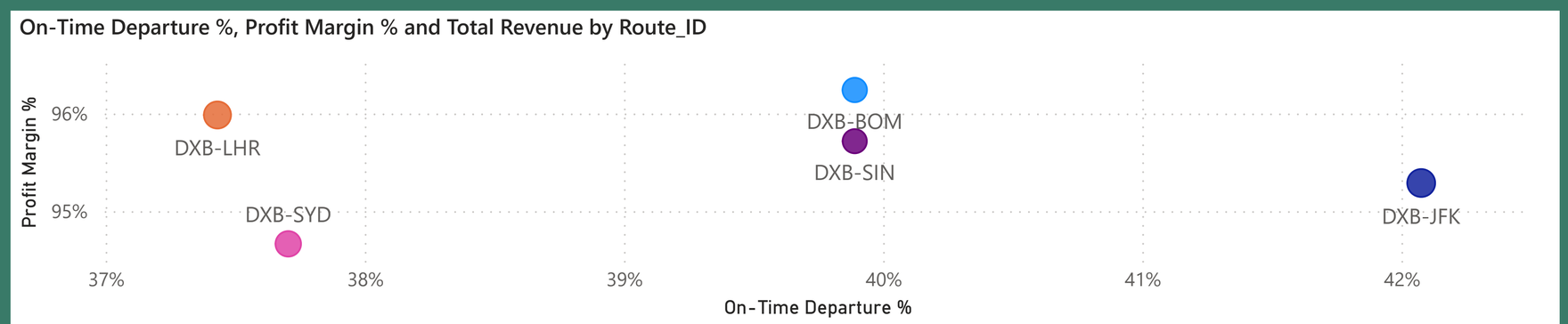




Operational Concentration:

Identifies routes or periods exhibiting disproportionate disruption, enabling targeted intervention rather than broad corrective measures.

Margin Impact Linkage



Linking Margin Impact:

Evaluates whether operational instability is materially compressing route-level profitability and elevating enterprise risk exposure.

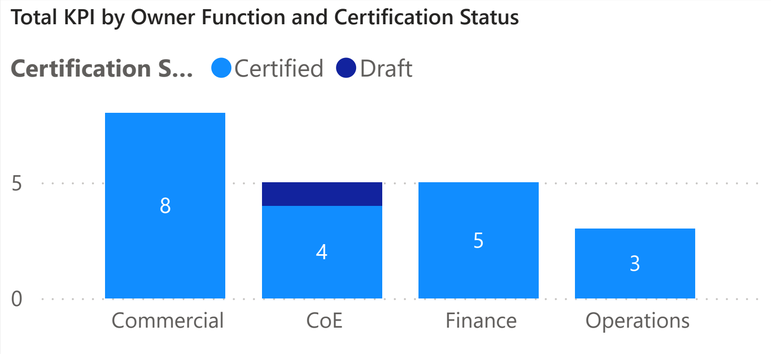
KPI GOVERNANCE & ANALYTICS CoE

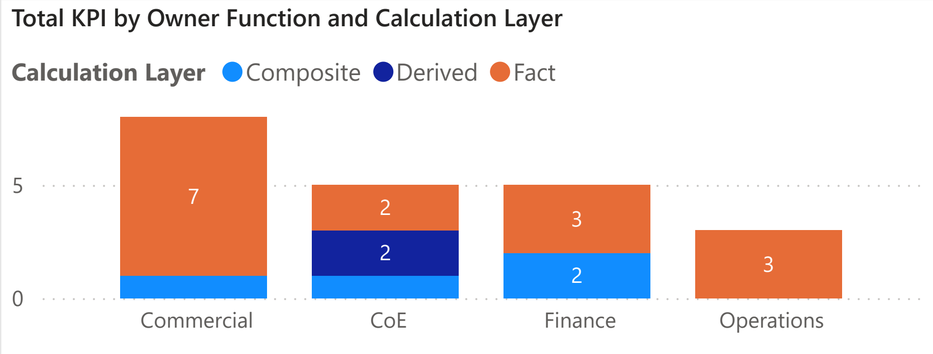
Governance Maturity Overview:

21	95.2%	100%	95%	95%
Total KPI	Certified KPI %	KPIs w/Assigned Ste...	KPIs w/Defined Quali...	SLA Compliance %

Governance & Coverage Snapshot:
Provides a consolidated view of governance maturity, certification coverage, and operational quality compliance across

Domain & Certification Control



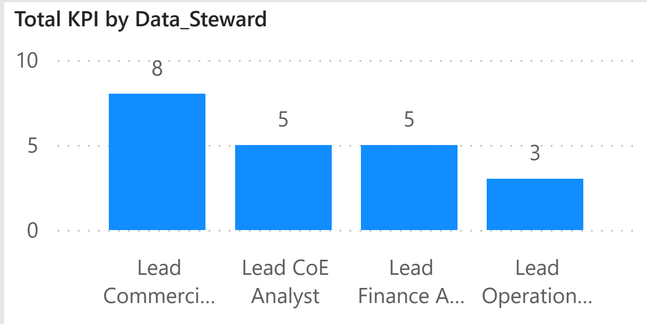


Domain & Certification Control:

Evaluates domain-level KPI ownership and certification discipline to ensure structural alignment and modeling transparency across the enterprise.

Ownership & Accountability Matrix

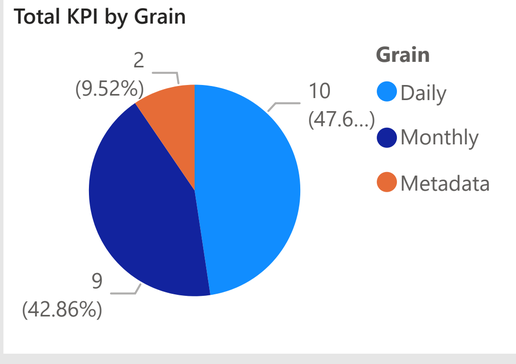
KPI Name	Owner Function	Data Steward	Certification Status	Calculation Layer
% Certified KPIs	CoE	Lead CoE Analyst	Certified	Derived
Advance Booking Wi...	Commercial	Lead Commercial ...	Certified	Fact
Ancillary Revenue %	Commercial	Lead Commercial ...	Certified	Fact
Cancellation Rate %	Operations	Lead Operations A...	Certified	Fact
Certified KPI Count	CoE	Lead CoE Analyst	Certified	Derived
Cost per Route	Finance	Lead Finance Anal...	Certified	Fact
Demand Momentu...	Commercial	Lead Commercial ...	Certified	Fact



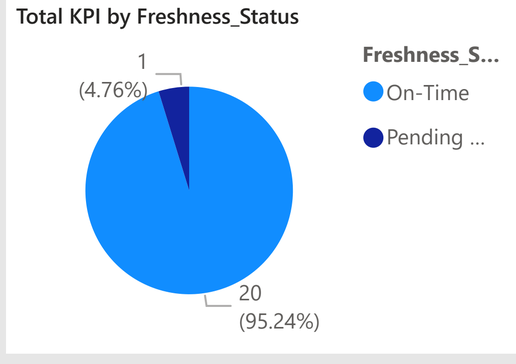
Ownership & Stewardship Accountability:

Surfaces executive accountability and operational stewardship responsibility to enforce semantic ownership and data governance discipline.

Data Quality, Lineage and Grain Integrity



KPI Name	Fact Source	Calc Layer	Fresh
% Certified KPIs	Dim_KPI_Definitions	Derived	As De
Advance Booking ...	Fact_Bookings	Fact	Daily
Ancillary Revenue...	Fact_Bookings	Fact	Daily
Cancellation Rate ...	Fact_Flights	Fact	Daily
Certified KPI Count	Dim_KPI_Definitions	Derived	As De
Cost per Route	Fact_Financials	Fact	Mont



Data Quality, Lineage and Grain Integrity:

Validates calculation grain alignment, refresh timeliness, and source lineage to prevent cross-domain ambiguity and reporting risk.